

Universality and a dimensional transcendence dichotomy for orthoscheme reflection groups

Vico Bonfioli

Abstract

The right-triangle orbit-growth universality of the companion papers is the two-dimensional case of a general phenomenon. We prove an abstract dichotomy: for any family of quotients of a common group, spherical growth is shape-independent through a horizon equal to half the shortest added relator, and first deviates exactly there. The orthoscheme (path-simplex) reflection groups realize it in every dimension — generic shapes are rigid, sharing one universal growth sequence, and each shape agrees up to a finite arithmetic horizon. Computing these universal sequences reveals a sharp *dimensional transcendence dichotomy*: the generic growth series is *rational* for $3 \leq n \leq 7$, with growth rate the explicit algebraic number $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$ (the spectral radius attached to the cycle C_{n+3}) and numerator $(1+z)^{\lfloor n/2 \rfloor + 1}$, whereas at $n = 2$ it is transcendental (OEIS A396406) with rate $\beta_2 = 1.4916\dots$ *strictly below* $r_2 = \varphi$. The plane is the unique dimension in which orbit growth is transcendental — equivalently, the unique dimension in which the growth rate falls short of the r_n envelope. This singles out the objects of the companion papers as the exceptional case of an otherwise rational family.

1 The abstract universality–deviation dichotomy

Let $G = \langle S \rangle$ be a group with finite symmetric generating set S and word length ℓ_G . For $N \trianglelefteq G$ write $Q = G/N$, $s_d(H) = \#\{h : \ell_H(h) = d\}$ for sphere sizes, and $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$.

Theorem 1 (Universality–deviation dichotomy). *With $n := \lceil K(N)/2 \rceil$,*

$$s_d(Q) = s_d(G) \quad (d < n), \quad s_n(Q) < s_n(G).$$

The first depth at which the quotient's growth falls below G 's is exactly $\lceil K(N)/2 \rceil$.

Proof. $\pi : G \rightarrow Q$ is surjective and 1-Lipschitz, so $s_d(Q) \leq s_d(G)$. If $d < n$ and $\pi g = \pi h$ with $g, h \in B_d(G)$, then $gh^{-1} \in N$ has length $\leq 2d < K(N)$, forcing $g = h$; the same bound rules out a strictly shorter representative, so π is length-preserving on $B_d(G)$ and $s_d(Q) = s_d(G)$ for all $d < n$, giving $|B_{n-1}(Q)| = |B_{n-1}(G)|$. Take $g_0 \in N$ with $\ell_G(g_0) = K(N)$ and split a geodesic word $g_0 = w_1 w_2$ with $\ell_G(w_1) = n$; both halves are geodesic, $\pi w_1 = \pi(w_2^{-1})$, and $w_1 \neq w_2^{-1}$, so two elements of $B_n(G)$ share a fiber and $|B_n(Q)| < |B_n(G)|$, whence $s_n(Q) < s_n(G)$. $\square$

Applied to a *family* $\{Q_T = G/N_T\}$ over a common G , Theorem 1 is a universality statement: all members share G 's growth through their horizon $n_T = \lceil K(N_T)/2 \rceil$, and members of larger $K(N_T)$ agree longer. For right triangles G is the generic (virtually lamplighter) group, $N_T = 2(t-1)\mu_T\mathbb{Z}[t^{\pm 1}]$, and n_T is the deviation-depth law of the companion paper; the theorem is the abstract skeleton of that result.

2 The orthoscheme family and generic rigidity

An n -orthoscheme (path simplex) has vertices $P_0, \dots, P_n$ with mutually orthogonal consecutive edges $P_{k-1}P_k = \ell_k e_k$ ($\ell_1 = 1$); its $n+1$ facet reflections $R_0, \dots, R_n$ generate $W_T \leq \text{Isom}(\mathbb{R}^n)$, the right-triangle side-reflection group when $n = 2$. Facets R_i, R_j with $|i - j| \geq 2$ are orthogonal, hence commute; consecutive facets meet at a dihedral angle that is, for generic legs, an irrational multiple of π . As for right triangles this rigidifies the generic group: its growth is independent of the (generic) shape. We measure geodesic growth $s_d(W_T)$ — the number of distinct isometries of word length d — by breadth-first search, deduplicating by the isometry (script `code/zeta_probe/orthoscheme_universality.py`; validated at $n = 2$ against A396406).

Proposition 2 (Generic universality). *For $2 \leq n \leq 7$, all tested generic n -orthoschemes share one geodesic-growth sequence $U^{(n)}$ (four shapes per dimension, agreement to the depths below), and the successive ratios converge to $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$.*

3 The dimensional transcendence dichotomy

Extracting the universal sequences and their generating functions $\mathcal{G}_n(z) = \sum_d s_d(U^{(n)})z^d$ gives the central finding.

Proposition 3 (Rationality for $3 \leq n \leq 7$). *For each $n \in \{3, 4, 5, 6, 7\}$ the universal sequence $U^{(n)}$ satisfies a linear recurrence, verified on all computed terms, with rational generating function*

n	$\mathcal{G}_n(z)$	$r_n = 1 + 2 \cos \frac{2\pi}{n+3}$	$[\mathbb{Q}(r_n) : \mathbb{Q}]$
3	$\frac{(1+z)^2}{1-2z}$	2	1
4	$\frac{(1+z)^3}{1-2z-z^2+z^3}$	$1 + 2 \cos \frac{2\pi}{7} = 2.24698\dots$	3
5	$\frac{(1+z)^3}{(1-z)(1-2z-z^2)}$	$1 + \sqrt{2}$	2
6	$\frac{(1+z)^4}{1-3z+3z^3}$	$1 + 2 \cos \frac{2\pi}{9} = 2.53209\dots$	3
7	$\frac{(1+z)^4}{(1-3z+z^2)(1+z-z^2)}$	$\varphi^2 = 2.61803\dots$	2

The numerator is $(1+z)^{\lfloor n/2 \rfloor + 1}$; the denominator's reciprocal is the characteristic polynomial of the growth, whose dominant root is r_n . In every case the rate equals r_n to full computed precision, and $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$ is the spectral radius shift attached to the cycle C_{n+3} (equivalently, $2 \cos \frac{2\pi}{n+3}$ is an eigenvalue of its adjacency operator).

Proposition 4 (Transcendence at $n = 2$). $U^{(2)}$ is OEIS A396406. It satisfies no linear recurrence (no relation up to order 8 on 17 terms) and is transcendental over $\mathbb{Q}(z)$: the relaxed companion is transcendental unconditionally, and the true series conditionally on the single amplitude estimate of the companion paper. Its growth rate is $\beta_2 = 1.4916177\dots$, strictly below $r_2 = \varphi = 1.6180\dots$

Propositions 3 and 4 together are the dichotomy.

Conjecture 5 (The plane is the unique transcendental dimension). *For every $n \geq 3$ the generic n -orthoscheme growth series is rational, with rate r_n ; only at $n = 2$ is it transcendental, and only there does the rate fall below r_n . Equivalently, the generic orthoscheme group is geodesically rational (indeed automatic) in every dimension except the plane, where it is virtually the lamplighter $\mathbb{Z} \wr \mathbb{Z}$.*

4 Discussion and status

Theorem 1 is unconditional and dimension-free: it reduces universality-then-deviation to a common ambient group and a shortest-relator length. For right triangles both inputs are supplied in closed form by the companion paper, making $n = 2$ a theorem with the deviation law and all; Propositions 2–4 establish the ingredients computationally for $2 \leq n \leq 7$, with the rational generating functions of Proposition 3 verified on every computed term (depths 13–16). What remains open — Conjecture 5 — is the structural identification of the generic n -orthoscheme group for $n \geq 3$: the higher analogue of the lamplighter theorem, which would turn the observed rationality into a theorem and pin the growth rate to r_n a priori.

The dichotomy recasts the companion papers. Their object — the transcendental growth of the planar right-triangle reflection group — is not an isolated curiosity but the *unique exceptional member* of a family that is otherwise rational in every dimension. The plane is singled out twice over, by the same feature: it is the only dimension whose generic orthoscheme group carries the lamplighter’s infinite lamp-space (hence transcendental growth), and correspondingly the only dimension whose growth rate β_2 falls short of the algebraic envelope $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$. Transcendence, here, is a two-dimensional phenomenon.